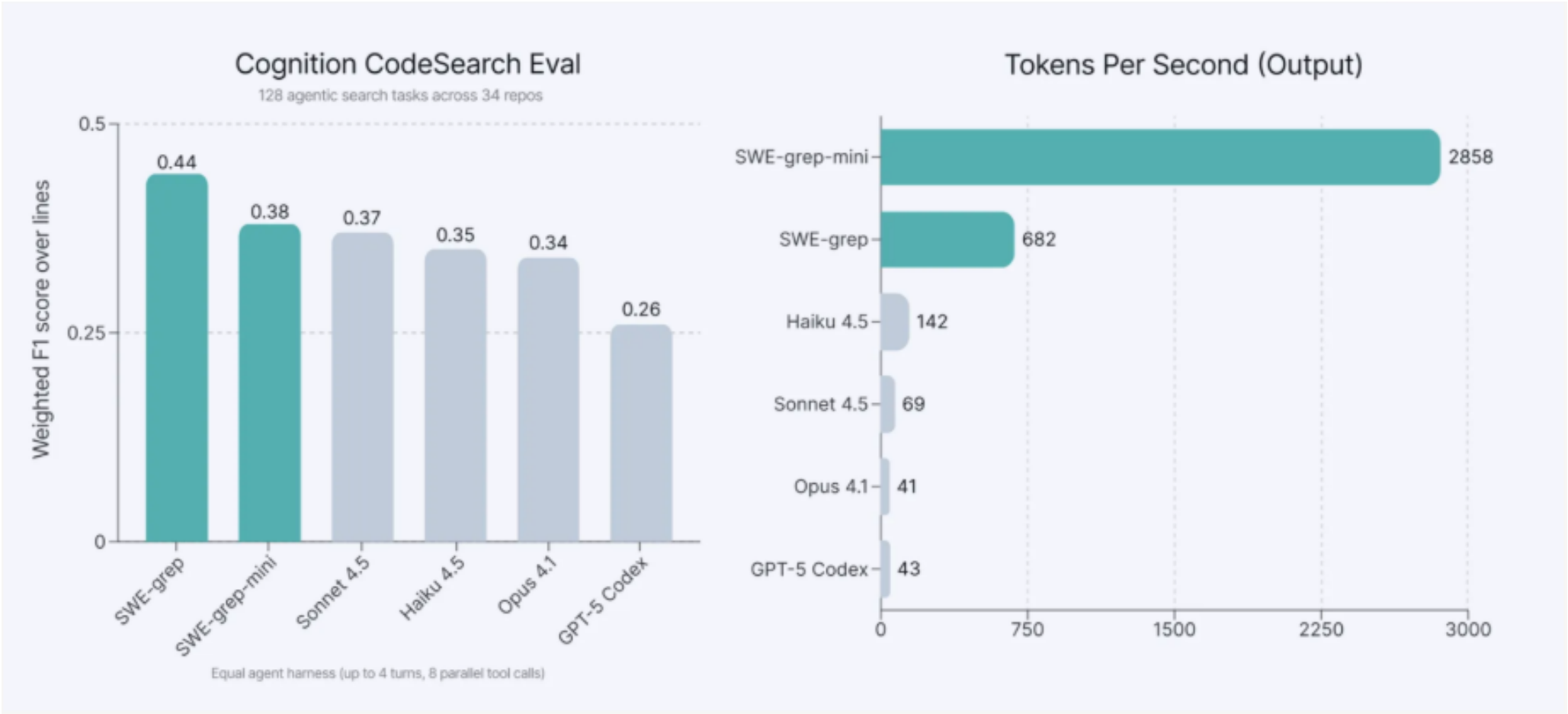
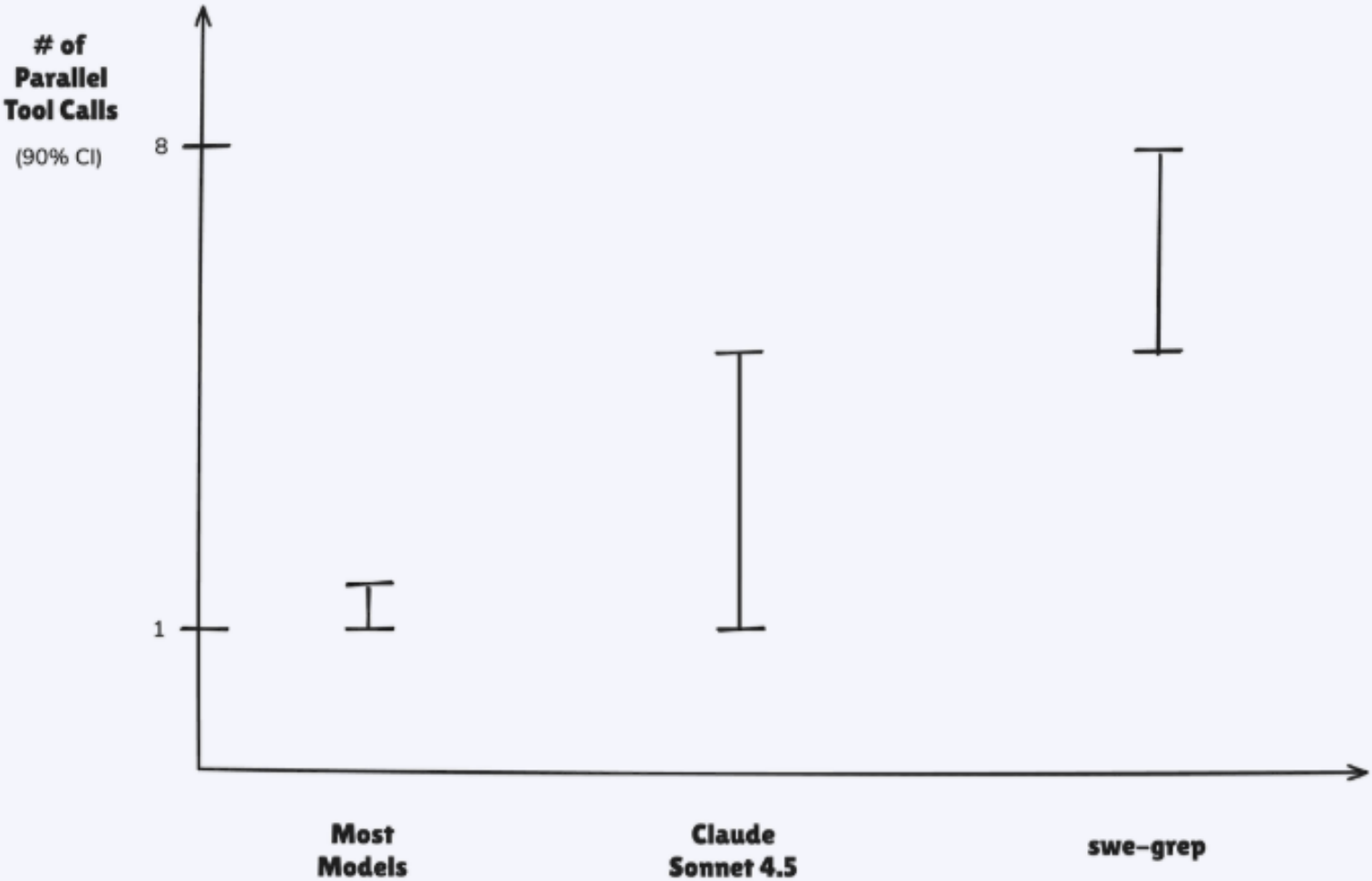


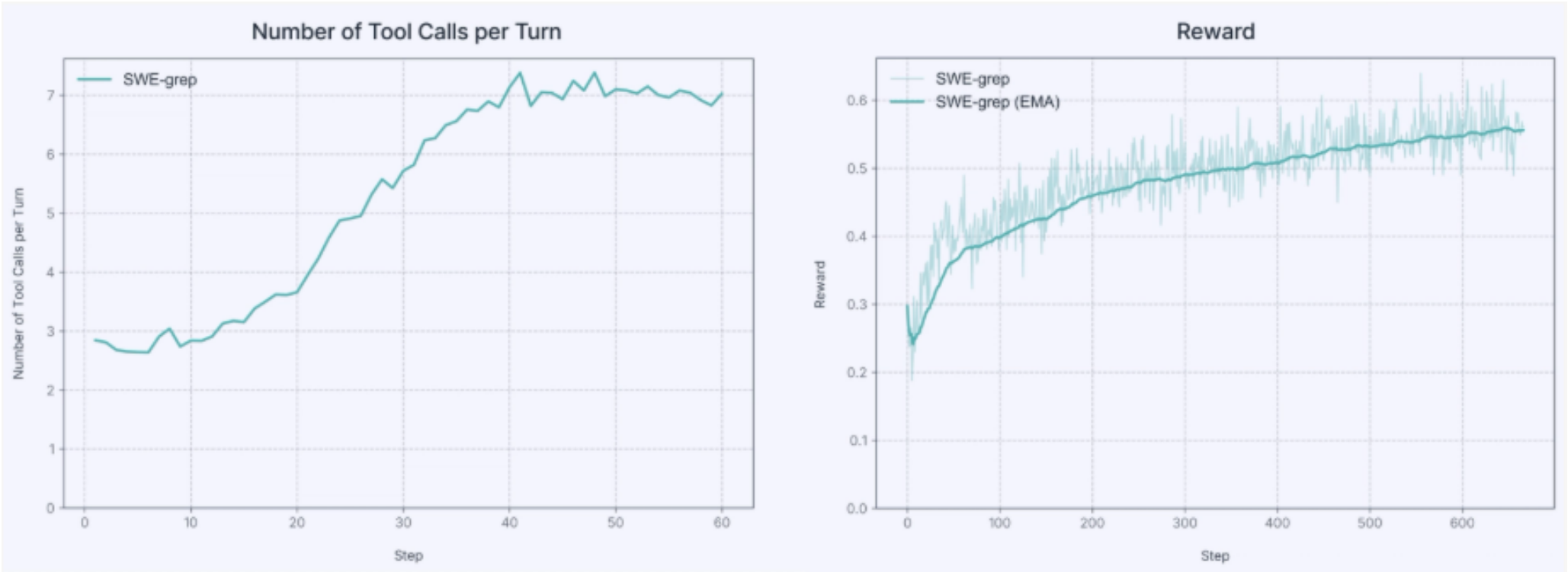


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Parallelism Design Goals

Numbers are guesstimates





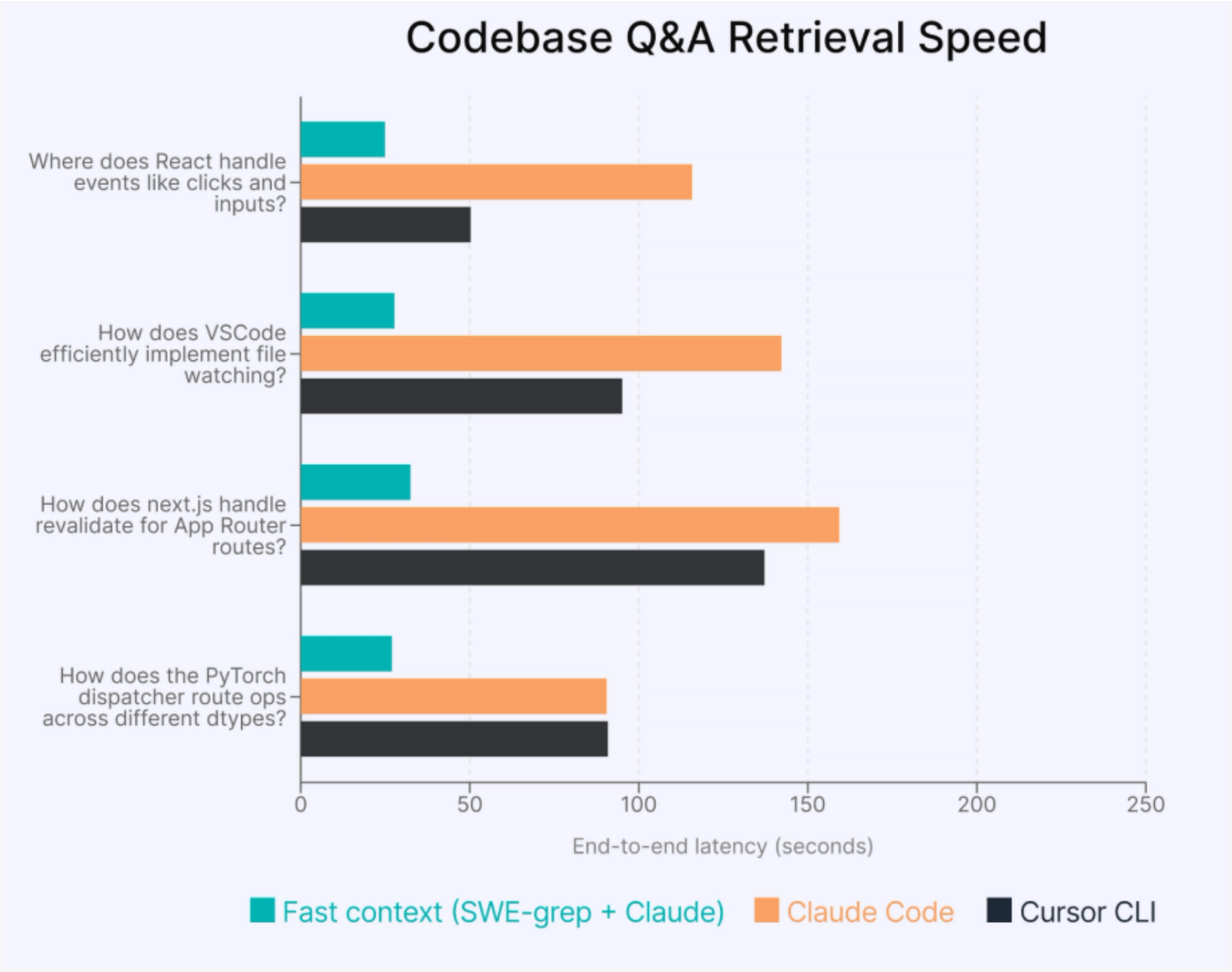
$$\mathbb{E}_{\tau \sim \pi_{\theta}} [R_{\tau} \nabla_{\theta} \log \pi_{\theta}(a|s)] = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_t R_{\tau} \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) \right]$$

$$\mathbb{E}_{\tau \sim \pi_{\text{sampler}}} \left[\frac{\pi_{\text{trainer}}(a|s)}{\pi_{\text{sampler}}(a|s)} R_{\tau} \nabla_{\theta} \log \pi_{\text{trainer}}(a|s) \right]$$

$$L(\theta) = -\frac{1}{GT_{\max}} \sum_{j=1}^G \left[\prod_{i=1}^{T_j} \frac{\pi_{\text{trainer}}(a_{j,i} | s_{j,i})}{\pi_{\text{sampler}}(a_{j,i} | s_{j,i})} \right]_{\nabla} \sum_{t=1}^{T_j} \frac{\pi_{\text{trainer}}(a_{j,t} | s_{j,t})}{[\pi_{\text{trainer}}(a_{j,t} | s_{j,t})]_{\nabla}} A_j$$

	F0.5 score over files	F0.5 score over lines	Tok/s (Output)	Latency (s)
SWE-grep	0.66	0.44	2858	2.79
SWE-grep-mini	0.58	0.38	682	1.82
Opus 4.1 (non-thinking)	0.54	0.34	41	64.41
Opus 4.1 (interleaved thinking)	0.55	0.35	40	86.26
Sonnet 4.5 (non-thinking)	0.59	0.37	69	35.90
Sonnet 4.5 (interleaved thinking)	0.59	0.38	65	59.06
Haiku 4.5	0.56	0.35	142	16.12
GPT-5-Codex	0.59	0.26	43	225.08
GPT-5 (High)	0.59	0.26	45	290.63
GPT-5 (Medium)	0.56	0.24	56	131.12
GPT-5 (Low)	0.54	0.22	50	71.05
GPT-5 (Minimal)	0.45	0.22	51	26.91
GPT-5 mini (Minimal)	0.36	0.15	61	15.23
Embedding (reranking, top-5)	0.47	0.29	N/A	2.79
Embedding (no reranking, top-5)	0.39	0.23	N/A	0.95
Embedding (reranking, top-10)	0.44	0.30	N/A	2.79
Embedding (no reranking, top-10)	0.39	0.25	N/A	0.95

	With Fast Context	Without Fast Context
Pass rate	63.90%	63.37%
Wall-clock time (mean)*	6 minutes, 6 seconds	7 minutes, 47 seconds
Total lines viewed (mean)	558	745
# of search file steps (mean)**	12.9	15.9



The Semi-Async Valley of Death

